Solution Architecture

Trip Planner — Supporting Project Document

Architecture Layers

Presentation layer: React/TypeScript/Vite. API layer: Python/FastAPI. AI integration: Groq. Persistence: Supabase PostgreSQL.

Frontend Responsibility

Collect inputs, send requests, display loading and error states, render generated sections and present saved-trip controls.

Backend Responsibility

Validate payloads, manage prompts and external API calls, read/write saved-trip data, and return safe responses.

AI Provider Responsibility

Generate text based on the prompt and provided preferences. It does not independently guarantee factual correctness.

Database Responsibility

Store trip fields and generated outputs so a user can revisit prior plans.

Deployment Considerations

Configure API routing, HTTPS, environment variables, allowed origins, authentication and database permissions for the deployment environment.